

A055 — Blind Topological Particle Spectrum Falsifier Chiral Covariance and Directional-Bias Certification

SST-Workbench

Version v0.4.0 — SST Falsifier Framework v1.0.4

1 Identification, scope, and status

A055 v0.4.0 is a frozen follow-up to the v0.3.1 signed-traveling-wave discovery analysis. It tests whether a one-way traveling-wave imbalance survives a circulation-relative gauge correction, exact representation metamorphic controls, the sealed spatial-resolution ladder, and both upstream Jacobian finite-difference steps. It does not modify upstream dynamics and does not claim independent physical replication.

2 Research question and objective

Does a representation-covariant circulation-relative directional bias persist across the frozen numerical controls of the anonymous three-component compound campaign?

3 Competing hypotheses and allowed conclusions

H_0 . No anonymous candidate retains the same circulation-relative directional-bias sign in at least two sectors across all registered resolutions and both Jacobian epsilon values.

H_1 . At least one anonymous candidate does. A PASS supports only a linear finite-model chirality signal; nonlinear confinement and particle identity remain unestablished.

4 Assumptions, domain of validity, and edge cases

The projected A054 Jacobian/mode basis is treated only as a finite-model diagnostic. Curve orientation and circulation sign form a representation pair: reversing both must not change a physical direction observable. v0.3.1 is discovery evidence from the same source family, not independent confirmation. A sector with fewer than three eligible modes cannot claim one-way bias.

5 Symbols, units, and dimensional consistency

Symbol	Meaning	Unit
$P_{jkc}^{\pm}$	component signed spatial-harmonic power	1
$\xi_{j,k}$	circulation-relative directional purity	1
$\eta_{j,k}$	harmonic participation	1
ζ_j	Kelvin-family fraction	1
B_s	sector vote bias	1

All decision observables are dimensionless.

6 Exact mathematical model and observables

$$P_{jkc}^{\pm} = |\hat{a}_{n,c}(\pm k)|^2 + |\hat{a}_{b,c}(\pm k)|^2. \quad (1)$$

$$\xi_{j,k} = \frac{\sum_c \text{sgn}(\Gamma_c)(P_{jkc}^+ - P_{jkc}^-)}{\sum_c (P_{jkc}^+ + P_{jkc}^-)}. \quad (2)$$

$$\eta_{j,k} = \frac{\sum_c (P_{jkc}^+ + P_{jkc}^-)}{P_{j,\perp}}, \quad B_s = \frac{N_+ - N_-}{N_+ + N_-}. \quad (3)$$

The key covariance identity is

$$\xi[C_c(-s), -\Gamma_c] = \xi[C_c(s), \Gamma_c]. \quad (4)$$

A mode is eligible at its dominant registered harmonic when $\zeta \geq 0.50$, $\eta \geq 0.50$, $|\xi| \geq 0.60$, and $|\Re \lambda|/|\Im \lambda| \leq 1.0$. A sector requires at least three eligible modes and $|B_s| \geq 0.50$.

Machine-readable science contract

Research question After correcting spatial propagation for filament orientation/circulation and enforcing representation covariance, does the one-way traveling-wave imbalance discovered in v0.3.1 persist as a resolution- and Jacobian-epsilon-stable property of anonymous three-component vortex compounds?

Objective Falsify a physical chirality interpretation of the v0.3.1 directional imbalance by replacing raw signed Fourier direction with a circulation-relative gauge-covariant observable, testing metamorphic representation invariance, and requiring recurrence over the sealed A054 spatial-resolution ladder and both preregistered Jacobian finite-difference steps.

H_0 H0: the v0.3.1 one-way imbalance is representation-, resolution-, or finite-difference-sensitive; no anonymous candidate retains a same-sign circulation-relative directional bias in at least two sectors across all frozen numerical controls.

H_1 H1: at least one anonymous candidate retains a same-sign circulation-relative directional bias in at least two sectors under all frozen representation, resolution, and finite-difference controls. This is a finite-model chirality signal only, not particle identification.

Registered equations

E1: Component-resolved signed powers

$$P_{jkc}^{\pm} = |\hat{a}_{n,c}(\pm k)|^2 + |\hat{a}_{b,c}(\pm k)|^2 \quad (5)$$

Dimensional check: Squared normalized projected amplitudes are dimensionless.

E2: Circulation-relative directional purity

$$\xi_{j,k} = \frac{\sum_c \text{sgn}(\Gamma_c)(P_{jkc}^+ - P_{jkc}^-)}{\sum_c (P_{jkc}^+ + P_{jkc}^-)} \quad (6)$$

Dimensional check: Ratio of equal-dimension powers; xi is dimensionless.

E3: Harmonic participation

$$\eta_{j,k} = \frac{\sum_c (P_{jkc}^+ + P_{jkc}^-)}{P_{j,\perp}} \quad (7)$$

Dimensional check: Ratio of powers; dimensionless.

E4: Sector directional vote bias

$$B_s = \frac{N_+ - N_-}{N_+ + N_-} \quad (8)$$

Dimensional check: Count ratio; dimensionless.

E5: Representation covariance identity

$$\xi[C_c(-s), -\Gamma_c] = \xi[C_c(s), \Gamma_c] \quad (9)$$

Dimensional check: Equality between dimensionless observables.

E6: Kelvin-family coefficient fraction

$$\zeta_j = \frac{\sum_{i \in \mathcal{K}} |v_{ij}|^2}{\sum_i |v_{ij}|^2} \quad (10)$$

Dimensional check: Ratio of squared normalized coordinates; dimensionless.

Registered code-to-mathematics steps

1. **S1** [G1; equations E5]: Verify sealed A054 source and framework-manifest-valid v0.3.1 discovery output without reading semantic private mappings.
2. **S2** [G2; equations E1]: Recompute projected Jacobians on the sealed spatial ladder and require fine-resolution eigenvalue-multiset agreement with the sealed spectrum.
3. **S3** [G3; equations E1, E2, E3, E4, E6]: For each positive-frequency mode compute component signed powers, Kelvin fraction, dominant harmonic, circulation-relative purity and sector vote bias.
4. **S4** [G2; equations E2, E5]: Apply gauge-equivalent cyclic-shift, rigid-rotation, component-permutation and orientation-reversal-plus-circulation-flip metamorphic controls to selected qualified modes.
5. **S5** [G5; equations E1]: Certify signed-harmonic powers with instance-local C++17/OpenMP FP64 against Python FP64.
6. **S6** [G8; equations E2, E4, E5]: Require same-sign sector bias above threshold at every sealed resolution and at both preregistered Jacobian epsilon values, recurring in at least two sectors of one anonymous candidate.
7. **S7** [G9; equations E4]: After commitment-verified reveal only, attach architecture/component semantics and compute paired contrasts using the frozen blind flags and biases.

7 Exact algorithm: equation-to-code mapping

- S1. Verify A054 and v0.3.1 provenance without semantic reveal.
- S2. Recompute projected Jacobians over the sealed resolution ladder and verify the fine sealed spectrum.
- S3. Compute component-resolved signed powers and circulation-relative mode direction.
- S4. Require a fine-resolution sector bias before interpreting chirality.

- S5. Apply cyclic-origin, rigid-rotation, component-permutation, and orientation-reversal plus circulation-flip metamorphic controls.
- S6. Certify the signed-power kernel with C++ FP64 parity.
- S7. Require same-sign sector bias over all registered resolutions and both Jacobian epsilon values, recurring in at least two sectors.
- S8. Reveal semantic labels only after commitment verification and compute paired contrasts.

8 Source, provenance, and independence contract

The authoritative numerical source is the sealed A054 v0.2.0-r2 compound campaign. A055 v0.3.1 is a frozen discovery baseline from the same simulation family. Neither is an independent experimental replicate. Missing, ambiguous, partial, or hash-invalid upstream data is UNRESOLVED.

9 Gate DAG and decision semantics

Gate	Question	PASS meaning
G0	Frozen protocol?	byte-identical
G1	Upstream provenance?	valid
G2	Reference numerics and representation covariance?	spectrum/selftest plus metamorphic error $\leq 10^{-8}$
G3	Fine circulation-relative bias?	at least one sector
G4	Independent confirmation?	disabled
G5	C++ parity?	relative L2 $\leq 10^{-10}$
G6	GPU?	disabled
G7	Independent confirmation?	disabled
G8	Resolution/epsilon recurrence?	at least one anonymous robust candidate
G9	Reveal integrity?	framework commitments verified

10 Numerical certification, convergence, and error budget

The A054 sealed ladder and Jacobian epsilon values are inherited unchanged. Fine-resolution eigenvalues must reproduce the sealed spectrum within assignment-relative error 10^{-8} . Representation metamorphic error is bounded by 10^{-8} . Directional recurrence uses the same fixed thresholds at every resolution and both epsilon values; no threshold is fit to the v0.3.1 effect size.

11 Backend authority and parity

Python/NumPy FP64 defines the circulation-relative analysis. Instance-local C++17/OpenMP FP64 certifies the underlying signed-harmonic powers against Python with relative L2 $\leq 10^{-10}$. Production uses the backend class recorded by the sealed A054 campaign; no silent fallback may close G5.

12 Blindness and reveal boundary

Blind computation uses anonymous candidate IDs, sector codes, sealed geometry/numerics, circulation signs, and frozen thresholds only. Architecture, component identity, and historical particle hypotheses remain private until reveal commitment verification.

13 Statistics and multiplicity

G8 is a recurrence/invariance gate, not a population-significance claim. The minimum of three eligible modes prevents a one-mode sector from generating a unit vote bias. Reveal-only architecture contrasts are paired by component assignment, provider stratum and sector when a matched control exists; pooled unpaired medians are not primary evidence.

14 Results insertion contract

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	5e62cad813e92513bf8703d8b3deb173b00a004c1f2bba2ab46176119a8e51b8
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	9734dba46bafc1ecdfe0ca84b9b1257702d012421e7e215766adbf44782984a2

Gate ledger

Gate	Status	Question
G0	PASS	Is the frozen framework protocol intact?
G1	PASS	Are the sealed A054 compound source and completed v0.3.1 discovery baseline
G2	PASS	Do sealed-spectrum recomputation, analytic covariance selftests, and real-mode
G3	PASS	Does any anonymous sector show the preregistered fine-resolution circulation-re
G4	NOT_APPLICABLE	Is there an independent held-out physical confirmation source?
G5	PASS	Does strict instance-local C++ FP64 certification reproduce Python signed-har
G6	NOT_APPLICABLE	Is optional GPU screening enabled and parity-qualified?
G7	NOT_APPLICABLE	Is an independent held-out physical confirmation source available?
G8	FAIL	Does any anonymous candidate retain a same-sign circulation-relative direction
G9	PASS	After blind gates, do reveal commitments verify and paired semantic contrasts

Backend manifest

```
{
  "a054_backend": {
    "actual": "cpp_pybind11_openmp",
    "sealed": "cpp_pybind11_openmp"
  },
  "framework_version": "1.0.4",
  "instance_cpp_metric": {
    "actual_backend": "openmp",
    "authority": "CERTIFICATION",
```

```

"build": {
  "actual_backend": "openmp",
  "artifact": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A055_Blin",
  "backend_info": {
    "backend": "openmp",
    "kernel": "biot_savart_plus_signed_harmonic_powers",
    "openmp_compiled": true,
    "openmp_max_threads": 20,
    "precision": "float64"
  },
  "compiler": "C:\\Program Files\\Microsoft Visual Studio\\2022\\Community\\VC\\Tools\\MSVC",
  "compiler_family": "msvc",
  "compiler_version": "Microsoft (R) C/C++ Optimizing Compiler Version 19.44.35228 for x64",
  "error": null,
  "fingerprint_sha256": "4e8ef07e9ad3a0491c517bfee26aeb95b1e0cf17b1042964f47cd2eec7e93b88",
  "flags": [
    "C++17",
    "O2/O3",
    "OpenMP"
  ],
  "python_abi": "cp314-win_amd64",
  "success": true,
  "toolset": "14.44.35207"
},
"parity": {
  "candidate_norm": 139.24626069569086,
  "pass": true,
  "reference_norm": 139.24626069569092,
  "relative_l2": 6.215789641365356e-16,
  "tolerance": 1e-10
},
"status": "PASS"
},
"schema": "A055-BACKEND-MANIFEST-040",
"summary": {
  "chiral_covariance": {
    "fine_biased_sector_count": 7,
    "max_metamorphic_abs_error": 6.661338147750939e-16,
    "mechanism_promoted_sector_count": 0,
    "robust_candidate_count": 0,
    "row_count": 288
  },
  "discovery_baseline": {
    "guard": "Discovery baseline only; raw signed direction is representation-dependent and i",
    "negative_direction_components": 26,
    "positive_direction_components": 238,
    "qualified_traveling_components": 264,
    "schema": "A055-V031-DISCOVERY-BASELINE-040",
    "sectors_with_both_directions": 16,
    "source_output": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A055"
  },
  "framework_version": "1.0.4",
  "interpretation": "A v0.4.0 robust candidate is evidence only for a representation-covarian

```

```

    "mode": "FULL",
    "schema": "A055-BLIND-SUMMARY-040"
  }
}

Source manifest

{
  "manifest_sha256": "a5cd68e6c5b919cdd41bc845a298bd62ed875afea5dfed7b87c10d118caaf06a",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "A054",
      "directory_manifest_sha256": "d3e5dc9bbc03bb4e59fe99602381dc2e6797ae4729837c5d46ac0b46777",
      "evidence_class": "simulation",
      "exists": true,
      "file_count": 170,
      "independence_family": "A054_FINITE_CORE_COMPOUND",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A054_nucleon",
      "provenance_family": "A054_V020_R2_SEALED",
      "registry_hint": {
        "catalog_id": "A054",
        "hits": []
      },
      "role": "primary_reanalysis_source",
      "source_id": "A054_COMPOUND",
      "version": "v0.2.0-r2"
    },
    {
      "catalog_id": "A055",
      "directory_manifest_sha256": "6beaf1187a1351c5451e15ef6cba6fcc5a1680e6ee47f911a9e264b681",
      "evidence_class": "simulation",
      "exists": true,
      "file_count": 22,
      "independence_family": "A055_V031_SAME_A054_FAMILY",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A055_Blind_T",
      "provenance_family": "A055_V031_FRAMEWORK_SEALED",
      "registry_hint": {
        "catalog_id": "A055",
        "hits": []
      },
      "role": "discovery_baseline",
      "source_id": "A055_V031_DISCOVERY",
      "version": "v0.3.1"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}

```

15 Allowed interpretation

G3 PASS with G8 FAIL means a finest-grid directional imbalance exists but is not numerically robust.
 G2 FAIL from the metamorphic subgate identifies a representation/numerical artifact. G8 PASS

supports a circulation-relative, representation-covariant, resolution/epsilon-stable linear chirality signal in the finite model only. Upstream spatial-convergence/RPO status is reported separately and remains required before nonlinear-mechanism language is allowed.

16 Reproducibility and exact commands

Canonical instance: 01_research/A_falsifiers/A055_Blind_Topological_Particle_Spectrum/A055-v0.4.0.

Canonical framework: 06_templates/SST_Falsifier_Framework/SST_Falsifier_Framework_v1.0.4.

Run `run_all.cmd full C:\workspace\projects\SST-Workbench`. The shipped instance is frozen; protocol mutation requires a new version.

References

- [1] SST-Workbench A054 v0.2.0-r2 sealed compound campaign.
- [2] SST-Workbench A055 v0.3.1 signed-travel discovery output.
- [3] SST Falsifier Framework v1.0.4 CANONICAL_FROZEN.